

DATABASE PROJECT REPORT

Due Date: Week 3,4 – Semester 1, Academic Year 2026–2027

Project ID & Title: #5: Healthcare Clinic & Telemedicine Portal

A. Project Identity

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Project Title: Healthcare Clinic & Telemedicine Portal

1. Introduction & Project Scope

1.1 System Objective

The Healthcare Clinic & Telemedicine Portal is a centralized database system designed to support comprehensive healthcare information management, clinical workflow coordination, and administrative operations for a clinic providing both traditional in-person consultations and telemedicine services. The system manages multi-role user authentication and access control, patient profiles, doctor information, doctor schedules, appointment booking, follow-up appointments, consultation sessions, medical histories, and digital prescriptions.

The primary objective is to provide a centralized and normalized database that maintains healthcare information in a consistent, secure, accurate, and reliable manner. The system must prevent scheduling conflicts and enforce strong relationships between doctors, schedules, appointments, consultation sessions, medical histories, and prescriptions.

The system applies Role-Based Access Control (RBAC) with three main roles:

ADMIN: responsible for system administration and clinic administrative operations, including managing user accounts, doctors, schedules, and appointments.

DOCTOR: responsible for clinical activities, including managing schedules, conducting consultation sessions, recording medical histories, issuing prescriptions, and creating follow-up appointments.

PATIENT: responsible for managing personal information, booking appointments, and accessing personal medical information.

General Practitioners and Specialists share common personal and professional information, including identification, contact information, and medical license numbers. Therefore, the database uses an EER specialization hierarchy with DOCTOR as the supertype and GENERAL_PRACTITIONER and SPECIALIST as two total and disjoint subtypes.

Major Objectives

1. To manage user credentials and enforce Role-Based Access Control (RBAC) across ADMIN, DOCTOR, and PATIENT roles.
2. To maintain centralized patient profiles containing demographic, contact, emergency contact, and baseline clinical information.
3. To manage doctor information and professional credentials.
4. To distinguish General Practitioners from Specialists using Total and Disjoint EER specialization.
5. To link medical specialists to their corresponding clinical specialties.
6. To manage doctors' working schedules and availability while preventing overlapping schedule slots.
7. To coordinate appointment bookings and support patient self-booking, administrator-assisted booking, and doctor-created follow-up appointments.
8. To manage consultation sessions and enforce that each consultation session is based on a valid appointment.
9. To record actual consultation start time, end time, duration, modality, and clinical notes.
10. To maintain longitudinal medical history and diagnosis records.
11. To manage prescriptions issued by attending doctors during valid clinical sessions.
12. To maintain data integrity and prevent inconsistent healthcare records.
13. To provide a rigorous conceptual database model for later logical design, SQL implementation, and Python web development.

System Scope

User & Access Management

Storing and managing user accounts, authentication data, account status, and role assignments for ADMIN, DOCTOR, and PATIENT users.

Patient Profile Management

Storing patient identification, contact information, emergency contacts, blood type, allergies, and chronic diseases.

Doctor Management

Storing doctor information, medical license numbers, professional credentials, and the EER specialization between General Practitioners and Specialists.

Schedule Management

Recording doctors' available working periods, duty schedules, and preventing overlapping schedule slots.

Appointment Management

Managing appointment booking, estimated consultation duration, appointment modality, operational status, self-booking, administrator-assisted booking, and doctor-created follow-up appointments.

Consultation Session Management

Recording actual clinical encounters, validating prior appointments, tracking start/end timestamps and actual duration, and supporting both in-person and telemedicine consultations. For virtual consultations, the system may store a meeting URL or virtual consultation reference to support remote patient-doctor interaction.

Medical History

Maintaining longitudinal diagnoses, symptoms, clinical observations, and consultation notes.

Digital Prescription

Recording prescriptions issued during valid consultation sessions, including medication, dosage, frequency, treatment duration, and special instructions.

Out of Scope

Hospital inpatient bed management.

Emergency trauma room operations.

Health insurance claim settlements.

Pharmaceutical warehouse inventory control.

Laboratory sample tracking.

Financial payment gateway processing.

1.2 Business Rules & Constraints

ID	Category	Description
BR1	User Account	Each user account must have a unique UserID.
BR2	User Account	Each user account must have a unique, non-null Username and a secure password hash.
BR3	User Account	Every account must be assigned exactly one role: ADMIN, DOCTOR, or PATIENT.
BR4	User Account	Every registered doctor and patient must be linked to exactly one valid USER_ACCOUNT.
BR5	User Account	An account cannot be deleted while active dependent operational records require it.
BR6	Patient Profile	Each patient must have a unique PatientID.
BR7	Patient Profile	Each patient must have basic identifying and contact information.

BR8	Patient Profile	A patient profile records baseline health information including blood type, allergies, chronic diseases, and emergency contact information.
BR9	Patient Profile	A patient can have zero, one, or many appointments over time.
BR10	Patient Profile	Each appointment belongs to exactly one patient.
BR11	Patient Profile	A patient can participate in zero, one, or many consultation sessions.
BR12	Patient Profile	A patient can have multiple medical history records and multiple prescriptions.
BR13	Doctor	Each doctor must have a unique DoctorID.
BR14	Doctor	Each doctor must have full name, contact information, and a unique LicenseNumber.
BR15	Doctor Specialization	Every doctor must be classified as either a General Practitioner or a Specialist.
BR16	Doctor Specialization	GENERAL_PRACTITIONER is a subtype of DOCTOR.
BR17	Doctor Specialization	SPECIALIST is a subtype of DOCTOR.
BR18	Doctor Specialization	Each Specialist must be associated with exactly one SPECIALTY.
BR19	Doctor Specialization	GENERAL_PRACTITIONER and SPECIALIST are disjoint; a doctor cannot belong to both.
BR20	Doctor Specialization	The specialization is total; every DOCTOR belongs to one of the two subtypes.
BR21	Schedule	A doctor can have zero, one, or many schedule records.
BR22	Schedule	Each schedule record belongs to exactly one doctor.
BR23	Schedule	Each schedule specifies ScheduleDate, StartTime, EndTime, and AvailabilityStatus.
BR24	Schedule	A doctor must not have overlapping working schedule periods on the same date.

BR2 5	Schedule	An appointment may only use an active and available working schedule.
BR2 6	Appointment	Each appointment must have a unique AppointmentID.
BR2 7	Appointment	Each appointment must be associated with exactly one patient and exactly one doctor.
BR2 8	Appointment	An appointment can be created by a patient, ADMIN, or assigned DOCTOR according to system permissions.
BR2 9	Appointment	Each appointment must specify date/time, estimated duration, and modality: In-person or Telemedicine.
BR3 0	Appointment	Status must be Scheduled, Checked-In, Completed, Cancelled, or No-show.
BR3 1	Appointment	A doctor-created follow-up appointment must reference the preceding appointment through a recursive relationship.
BR3 2	Appointment	If a patient does not attend, the appointment status must be updated to No-show.
BR3 3	Appointment	Cancelled and No-show appointments remain recorded in appointment history.
BR3 4	Appointment	A doctor must not have overlapping active appointments.
BR3 5	Appointment	Cancelled and No-show appointments are not treated as active appointments for conflict checking.
BR3 6	Consultation	Each consultation session must have a unique SessionID.
BR3 7	Consultation	A consultation session can only exist for a valid appointment.
BR3 8	Consultation	Each consultation session must be linked to exactly one appointment; patient and doctor attributes are implicitly accessed through the associated appointment.
BR3 9	Consultation	Each session records exact start time, end time, and computed actual duration in minutes.
BR4 0	Consultation	Each session records clinical notes and whether it is In-person or Virtual.

BR4 1	Medical History	A patient may accumulate multiple medical history records.
BR4 2	Medical History	Each medical history record is linked to the consultation session in which the diagnosis was established, deriving patient information directly from the parent session.
BR4 3	Medical History	A record contains diagnosis, symptoms, record date, and progress notes.
BR4 4	Medical History	Previous medical history records are permanent clinical archives and should not be deleted.
BR4 5	Prescription	Each prescription must have a unique PrescriptionID.
BR4 6	Prescription	A prescription must be issued during a valid consultation session, deriving patient and attending doctor details directly from that session.
BR4 7	Prescription	The attending doctor and patient assigned to a prescription are determined strictly by the parent consultation session.
BR4 8	Prescription	Each prescription must contain one or more PRESCRIPTION_ITEM records.
BR4 9	Prescription	Each prescription item must reference a valid MEDICATION and define dosage, frequency, duration, and special instructions.
BR5 0	Cross-Entity Integrity	APPOINTMENT.DoctorID must match the DoctorID of the selected DOCTOR_SCHEDULE.
BR5 1	Cross-Entity Integrity	AppointmentDateTime must occur on the selected ScheduleDate and fall within its StartTime–EndTime interval.
BR5 2	Workflow Integrity	MEDICAL_HISTORY can only be created when the related APPOINTMENT has Status = Completed.
BR5 3	Workflow Integrity	PRESCRIPTION can only be created when the related APPOINTMENT has Status = Completed.
BR5 4	Data Inheritance	Patient and Doctor information for prescriptions and medical histories are implicitly derived from the parent Consultation Session to maintain normalization and avoid data redundancy.

BR57	Telemedicine Session	A telemedicine appointment must use a Virtual consultation session when the consultation is conducted remotely.
BR58	Virtual Meeting Information	A Virtual consultation session may store a meeting URL or virtual consultation reference required for remote patient-doctor interaction.

ID	Category	Description
IC1	Integrity	Entity Integrity: Every primary entity must have a unique, non-null primary key.
IC2	Integrity	Referential Integrity: Every foreign key must reference an existing parent key.
IC3	Integrity	Domain Integrity: Role, AppointmentType, Status, AvailabilityStatus, dates, and timestamps must conform to defined domains.
IC4	Integrity	Uniqueness Constraints: Username, PatientID, DoctorID, and LicenseNumber must be unique where required.
IC5	Integrity	Specialization Integrity: Every doctor belongs to exactly one of the two total and disjoint subtypes.
IC6	Integrity	Scheduling Integrity: A doctor cannot have overlapping working schedules or overlapping active appointments.
IC7	Integrity	Workflow Integrity: A consultation session cannot exist without a valid appointment; medical history and prescriptions require a completed appointment.
IC8	Integrity	Temporal Integrity: AppointmentDateTime must match the selected schedule date and time interval; consultation start/end times must be logically ordered.

IC9	Integrity	Cross-Entity Consistency: The appointment doctor must own the selected schedule.
IC10	Integrity	Clinical Record Consistency: Prescription and medical history records are linked solely to the consultation session, ensuring normalized references without redundant patient or doctor attributes.
IC11	Integrity	Telemedicine Integrity: When an appointment is conducted as Telemedicine, the corresponding consultation session must use SessionType = Virtual.

The conceptual model represents the main entities, attributes, relationships, cardinalities, and specialization required by the Healthcare Clinic & Telemedicine Portal.

Main Entities

Entity	Keys (PK/FK)	Attributes
USER_ACCOUNT	PK: UserID	Username, PasswordHash, Role (ADMIN, DOCTOR, PATIENT), AccountStatus, CreatedAt
PATIENT	PK: PatientID, FK: UserID	FullName, DateOfBirth, Gender, Phone, Email, Address, BloodType, Allergies, ChronicDiseases, EmergencyContactName, EmergencyContactPhone
DOCTOR	PK: DoctorID, FK: UserID	FullName, Phone, Email, LicenseNumber
GENERAL_PRACTITIONER	PK/FK: DoctorID	Subtype of DOCTOR
SPECIALIST	PK/FK: DoctorID, FK: SpecialtyID	Subtype of DOCTOR
SPECIALTY	PK: SpecialtyID	SpecialtyName, Description
DOCTOR_SCHEDULE	PK: ScheduleID, FK: DoctorID	ScheduleDate, StartTime, EndTime, AvailabilityStatus
APPOINTMENT	PK: AppointmentID, FK: PatientID, DoctorID, ScheduleID, BookedByUserID, FollowUpFromApptID (Recursive)	AppointmentDateTime, EstimatedDurationMinutes, AppointmentType (In-person, Telemedicine), Status (Scheduled, Checked-In,

		Completed, Cancelled, No-show), Reason
CONSULTATION_SESSION	PK: SessionID, FK: AppointmentID (Unique)	StartTime, EndTime, ActualDurationMinutes, SessionType (In-person, Virtual), MeetingURL, DiagnosisNotes
MEDICAL_HISTORY	PK: MedicalHistoryID, FK: SessionID	RecordDate, Diagnosis, Symptoms, ProgressNotes
PRESCRIPTION	PK: PrescriptionID, FK: SessionID	DiagnosisICD, PrescriptionDate, Instructions
PRESCRIPTION_ITEM	PK: PrescriptionItemID, FK: PrescriptionID, MedicationID	Dosage, Frequency, Duration, SpecialInstructions
MEDICATION	PK: MedicationID	MedicationName, Description

The system uses USER_ACCOUNT as the central entity for Role-Based Access Control (RBAC). Each account is assigned exactly one role: ADMIN, DOCTOR, or PATIENT.

ADMIN

Manage user accounts.

Manage doctor information.

Manage doctor schedules.

Create appointments on behalf of patients.

Adjust or cancel appointments.

Manage administrative information.

No separate ADMIN entity is required because the administrative role is represented through the Role attribute of USER_ACCOUNT.

DOCTOR

Manage working schedules.

View assigned appointments.

Conduct consultation sessions.

Record medical history.

Issue prescriptions.

Create follow-up appointments.

PATIENT

Manage personal information.

Book appointments.

View appointments.

Participate in telemedicine consultations.

View personal medical history.

View prescriptions issued to them.

Relationships: USER_ACCOUNT → DOCTOR: 1 : 0..1; USER_ACCOUNT → PATIENT: 1 : 0..1.

Every DOCTOR and PATIENT profile must be associated with exactly one USER_ACCOUNT.

Major Relationships

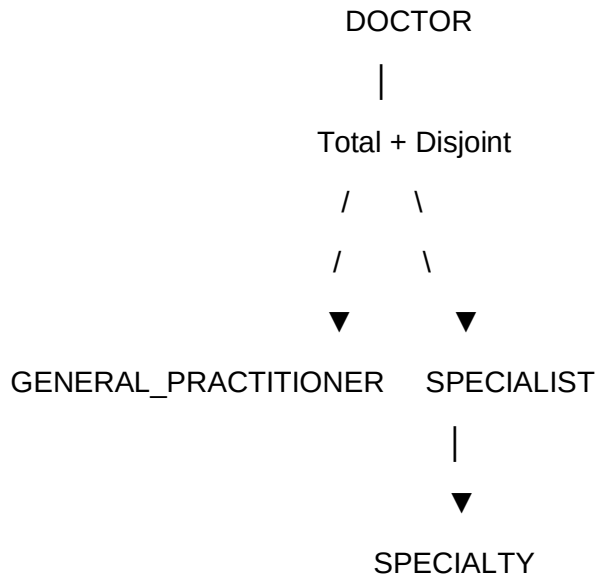
Entities	Relationship / Cardinality
USER_ACCOUNT - DOCTOR	1 : 0..1
USER_ACCOUNT - PATIENT	1 : 0..1
PATIENT - APPOINTMENT	1 : N
DOCTOR - APPOINTMENT	1 : N
USER_ACCOUNT - APPOINTMENT	1 : N – identifies the user who creates the appointment
DOCTOR_SCHEDULE - APPOINTMENT	1 : N – subject to schedule/time integrity constraints
APPOINTMENT - APPOINTMENT	1 : 0..1 – recursive follow-up relationship
APPOINTMENT - CONSULTATION_SESSION	1 : 0..1
CONSULTATION_SESSION - MEDICAL_HISTORY	1 : 0..N – derives patient context via consultation session
CONSULTATION_SESSION - PRESCRIPTION	1 : 0..N – derives patient and doctor context via consultation session
PRESCRIPTION - PRESCRIPTION_ITEM	1 : N

MEDICATION - PRESCRIPTION_ITEM	1 : N
SPECIALTY - SPECIALIST	1 : N
DOCTOR - DOCTOR_SCHEDULE	1 : N

EER Specialization

DOCTOR Specialization

The **DOCTOR** entity is specialized into two subtypes:



The specialization is Total and Disjoint. Total specialization means every **DOCTOR** must belong to one subtype. Disjoint specialization means a doctor cannot simultaneously be both a **GENERAL_PRACTITIONER** and a **SPECIALIST**.

Conceptual ER/EER Diagram

HEALTHCARE CLINIC & TELEMEDICINE PORTAL
EER DIAGRAM

